

# Network or Control? A Validated Spectral Bridge for Inverter-Dominated Grids

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**Abstract**—Industrial grids and microgrids are replacing synchronous machines with inverter-based resources, so damping is no longer supplied mainly by rotating-machine inertia. Much of it now lives in controller states: phase-locked loops, exciters, governors, and inverter loops. This shift breaks a common shortcut in weak-node and weak-link screening. A static reduced model that assigns damping to a fixed nodal coefficient can report a benign electromechanical margin while the actual least-damped pole belongs to a controller family.

This poster presents a controller-aware spectral bridge for that case. Instead of collapsing condensed controller dynamics into a scalar nodal damping coefficient, the bridge retains them as a frequency-dependent self-energy inside a rational Schur nonlinear eigenvalue problem. On IEEE 39-bus ANDES diagnostic cases, the scalar bridge fails because the packaged GENROU.D coefficients are zero, while the full eigensolve still shows modal damping supplied by controller and auxiliary states. The controller-aware bridge recovers the full ANDES critical poles in no-PSS, base, and Mix60 cases. The worst observed errors are  $8.47 \times 10^{-11}\%$  in frequency and  $2.48 \times 10^{-9}\%$  in damping. In Mix60, the least-damped mode is not the 1.49 Hz electromechanical mode found by scalar and fixed-point reductions. It is a 13.7417 Hz PLL-family mode with damping ratio  $\zeta = 0.12357$  and control participation  $\pi_c = 0.853$ .

**Index Terms**—inverter-based resources, small-signal stability, damping margin, phase-locked loop, weak-node identification, weak-link analysis, power-system planning

## I. MOTIVATION AND REDUCED OBJECT

Partition the linearized ANDES model into retained electrical or network-interface states and condensed controller states. Eliminating the condensed block gives a rational reduced object,

$$F(s) = A_{rr} - sI + A_{rc}(sI - A_{cc})^{-1}A_{cr},$$

or, in second-order notation, a network operator corrected by a frequency-dependent self-energy  $\Sigma(s)$ . This is the object that should be screened. The static limit loses the frequency dependence that carries the control poles, and it is not reliable when  $A_{cc}$  contains integral controller states or is ill-conditioned near zero frequency.

The tools are classical: Schur complements, rational eigenvalue problems, contour integration, and modal participation. The contribution is the formulation and validation of this object as a planning bridge. If a reduced model is going to speak about damping margins, it must retain the controller dependence that supplies those margins. Otherwise it can track the wrong pole.

## II. VALIDATION RESULT

The bridge audit compares rational reduced poles with the full ANDES eigensolve in three diagnostic cases. In no-PSS

and base cases, the critical modes remain network-family. In Mix60, where 60% of synchronous generation is replaced by a mix of grid-following and grid-forming inverter models, the critical mode changes family. Scalar and fixed-point reductions continue to look near the low-frequency electromechanical mode. The rational bridge finds the PLL-family pole at 13.7417 Hz.

This result changes the engineering question. The issue is no longer only where the grid is weak. It is what makes the weak element weak. A low-frequency network mode and a PLL-family mode ask for different interventions. A ranking that cannot separate those families can point to the wrong class of action.

## III. WEAK NODES, WEAK LINKS, AND SCOPE

The poster uses the bridge as a classifier before using it as a ranking tool. Network-family weak nodes are candidates for topology reinforcement. Control-family weak nodes should be treated as tuning problems first. Control-resonant weak links require more care: reinforcing the line may pull a network mode toward a condensed controller pole and reduce damping instead of improving it.

The present evidence supports the bridge and one bridge-enabled mechanism audit. It does not yet support a full planning claim across calibrated operating ranges. The ANDES cases are diagnostic, and the controller parameters are generic. The strict resonant mechanism appears in a minority of audited records, 40/348, and none of those resonant network modes becomes the global critical pole. The claim is limited by design: the bridge identifies a mechanism that scalar reductions miss, not a universal rule for every reinforcement reversal.

## IV. POSTER HOOK AND TAKEAWAY

Not every weak node is weak for the same reason. Some need copper. Some need control. The poster centers this point with the Mix60 complex-plane view: the mode that classical reductions find near 1.49 Hz is not the least-damped mode. The critical pole is the 13.7417 Hz PLL-family mode.